

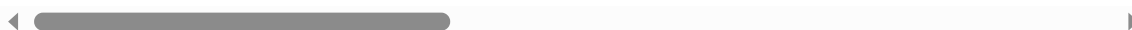
```
In [19]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [20]: df=pd.read_csv('E_Commerce.csv')
df      #here df stands for data file
```

```
Out[20]:
```

	order_id	order_date	state	zone	category	brand_type	customer_c
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	
...	...	...	...	...	...	...	...
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

30600 rows × 16 columns



```
In [21]: from scipy.stats import binom
```

```
In [22]: len(df)
```

```
Out[22]: 30600
```

```
In [23]: df['brand_type'].value_counts()
```

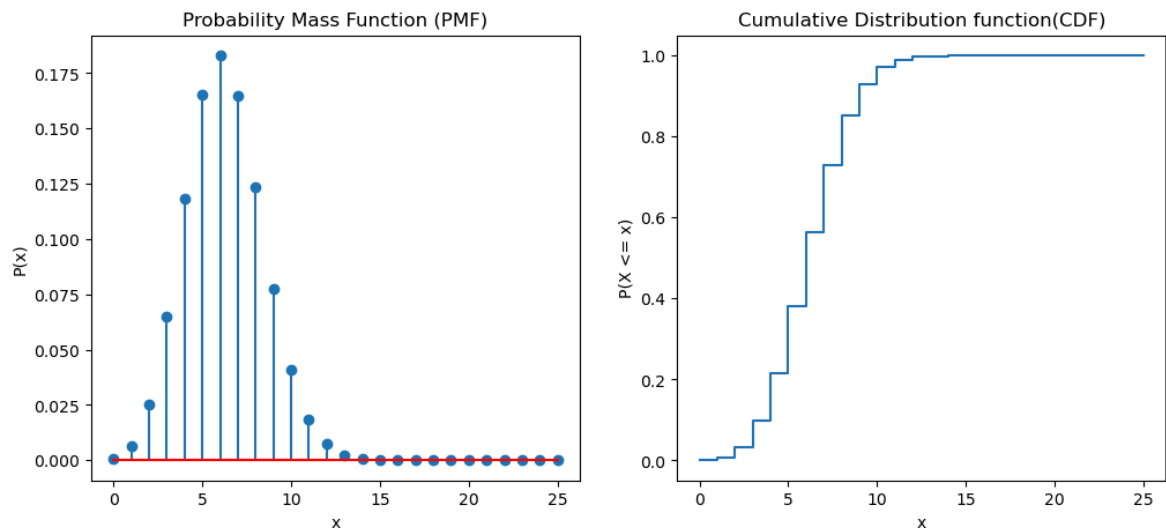
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Out[23]: brand_type
Mass      22970
Premium   7630
Name: count, dtype: int64
```

What is the probability that at least 10 out of 25 randomly selected brand types are premium? Graph PDF and CDF

```
In [34]: total=len(df)
p_premium = (df['brand_type'] == 'Premium').sum()/total
n = 25
k = 9
p_at_least_10 = 1 - binom.cdf(k, n, p_premium)
print(f"Estimated Premium probability : {p_premium:.3f}")
print(f"P(at least 10 Premium out of 25) = {p_at_least_10:.4f}")
x = np.arange(0, n+1)
pmf = binom.pmf(x, n, p_premium)
cdf = binom.cdf(x, n, p_premium)
plt.figure(figsize=(12, 5))
#PMF
plt.subplot(1, 2, 1)
plt.stem(x, pmf, basefmt="r-")
plt.title("Probability Mass Function (PMF)")
plt.xlabel("x")
plt.ylabel("P(x)")
#CDF
plt.subplot(1, 2, 2)
plt.step(x, cdf, where='post')
plt.title("Cumulative Distribution function(CDF)")
plt.xlabel("x")
plt.ylabel("P(X <= x)")
plt.show()
```

Estimated Premium probability : 0.249

P(at least 10 Premium out of 25) = 0.0702



In []: